

Davide Macario

dav.macario@gmail.com | blog.dmhomed.com | github.com/davmacario | www.linkedin.com/in/davide-macario-b872b4225/

Skills

Technical	Python, Golang, Terraform, AWS, Linux, Bash, Kubernetes, Docker, CI-CD tooling (Git, GitHub Workflows, GitLab CI), Zero-trust, Ansible
Languages	Italian (native), English (C1 – IELTS 8.0), French (basic), Dutch (basic)
Interests	ML/AI applications, Open-Source Software, MLOps, Self-Hosting/Homelab, Distributed Computing, Software Architectures, Edge Computing, IoT, Linux

Experience

Vitestro B.V., DevOps Engineer | Utrecht, NL Mar 2026 – Current

- Lead effort to achieve integration of product with hospital's Laboratory Information Systems (LIS); deployed and maintained integration engine hosted on **AWS** (ECS Fargate), including centralized monitoring platform.
- Implemented solution for **remote data upload** (including telemetry, and sessions) from the device to **AWS**, including the use of **Fluent Bit** and custom tooling for specific data types, enabling near-real-time retrieval of data.
- Supported the other R&D teams with **internal tooling**, including data migrations, package registries, container registries, container image development.

Vitestro B.V., Entry-Level DevOps Engineer | Utrecht, NL Sep 2024 – Feb 2026

- Member of Platform & Connectivity team working on integrating the world's first **Autonomous Robotic Phlebotomy Device** with local and public cloud.
- Engineered a full **GitLab CI** ecosystem on **AWS**, including autoscaled runners, Docker registry mirror, repository caches; achieved costs reduction compared to existing setup.
- Contributed to the development and maintenance of custom Embedded Linux distribution (**Yocto, Bitbake**).

University of Illinois at Chicago, Graduate Research Assistant | Illinois, USA Jan 2024 – Jun 2024

- Research work focusing on distributed computing applied to **Machine Learning** and **Generative AI** models.
- Designed and implemented a framework for deploying state-of-the-art LLMs on networks of constrained edge devices.
- Won **Best Paper Award** at IEEE LANMAN 2025 for the work on “A Model-Distributed Inference Approach for Large Language Models at the Edge”.

Education

University of Illinois at Chicago, MS in Electrical and Computer Engineering | IL, USA Jul 2024

Double-degree program in collaboration between Politecnico di Torino and University of Illinois at Chicago.

Politecnico di Torino, MS in ICT for Smart Societies (Computer Engineering) | Torino, Italy Jul 2024

M.Sc. Thesis: “A Model-Distributed Inference Approach for Large Language Models at the Edge”.

Politecnico di Torino, BS in Electronics and Communications Engineering | Torino, Italy Sep 2022

Associations: IEEE Eta-Kappa-Nu Mu-Nu chapter member (honor society – Politecnico di Torino).

Projects

DMHomed – Kubernetes-powered Homelab Nov 2025 – Current

- Bare-metal **Kubernetes** (K3s) cluster running in homelab, running both internal (VPN) and public-facing services.
- **GitOps** setup via ArgoCD, centralized **monitoring and observability** (Prometheus), **Ansible** for provisioning and server upgrades, **CloudNativePG** for databases, **Tailscale** as VPN, and **Cloudflare Tunnels** for public services.
- Migrated from existing Docker setup without data loss.
- <https://github.com/davmacario/dmhomed-infra>

A Model-Distributed Inference for Large Language Models at the Edge | M.Sc. Thesis Jan 2024 – Jun 2024

- Implementation of **pipeline parallelism** for **Large Language Models**, allowing to run LLM inference on a network of edge devices (Nvidia Jetson TX2).
- Employing **distributed computing** techniques to achieve increased generation rates by minimizing idle nodes when serving multiple user queries at the same time.
- Ability to run the **GPT-2**, **Llama 3**, and **Mistral** model families achieving usable throughput.
- <https://github.com/davmacario/MDI-LLM>

- FREISA** | OpenCV AI Competition 2023 (*Grand prize winner*)

Jul 2023 – Dec 2023
- “*Four-legged Robot Ensuring Intelligent Sprinkler Automation*”.
 - Computer vision-enabled quadruped robot able to detect plants, approach them and decide whether to sprinkle water on them based on the condition of the leaves.
 - Focused on developing the computer vision module using an OAK-D Lite camera, **YOLOv8**, **OpenCV**, and **DepthAI**, providing fast APIs to the motion controller (ROS 2), and running fully on-board.
 - <https://github.com/B-AROL-0/FREISA>

Certifications	
Software Architecture for Developers – by Simon Brown (Trifork)	May 2026
HashiCorp: Terraform Associate (003)	Aug 2025

Achievements	
Best Paper Award – IEEE LANMAN 2025, Lille, France	Jul 2025
Alta Scuola Politecnica (ASP) – honor programme for top 60 students during M.Sc.	Sep 2022 – Jul 2024
TOP-UIC Scholarship – awarded on academic performance for the double-degree programme	May 2023
Percorso Giovani Talenti – honor programme for top students during B.Sc.	Sep 2019 – Jul 2022